

Zhang Jiahao

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Research Focus

Explainable AI as a bidirectional, interactive system: I study *(i) understanding*—mechanistic, concept-level explanations of model decisions—and *(ii) intervention*—human-guided control to reliably steer model behavior. I validate these ideas through **AI-for-Science** applications, focusing on faithfulness, reliability, and downstream scientific utility.

Research Interests

- **Mechanistic Interpretability for Transformers:** concept/neuron-path discovery, circuit discovery, tool-calling mechanisms, and cross-layer information flow tracing.
- **Human-in-the-Loop Intervention & Control:** test-time steering, concept-based interventions, robustness/reliability under weak or indirect supervision.
- **Data-centric and Agentic LLMs:** supervised fine-tuning data recipe search, agentic tool-use mechanisms, and reliable model behavior under complex prompts.
- **XAI for Scientific Discovery (AI4Science):** interpretable and controllable modeling for protein generation, brine chemistry, and biomedical/spatial-omics analysis.

Education

- **Ph.D. in Machine Learning, MBZUAI** Aug. 2025 – Present
Abu Dhabi, UAE
Supervisor: Lijie Hu
Theme: interactive XAI, mechanistic interpretability, and AI4Science.
- **B.Eng. in Electronic Information, HUST (985)** Sept. 2021 – Jun. 2025
Wuhan, China
GPA: 86/100
- **Visiting Student, UC Berkeley** Jan. 2024 – May 2024
Berkeley, CA
GPA: 3.57/4.0

Publications

* denotes equal contribution (co-first authors). † denotes corresponding author(s).

Accepted / Published

- **Jiahao Zhang***, Zeqing Zhang*, Di Wang, Lijie Hu[†]. *Controlling Repetition in Protein Language Models*. ICLR 2026 (accepted), Poster.
XAI/Intervention angle: diagnosed repetition failure modes in protein language models and proposed an inference-time steering method, UCCS, that reduces degeneracy while preserving downstream foldability measured by AlphaFold confidence.
- Peng Cui*, **Jiahao Zhang***, Lijie Hu[†]. *Bayesian Gated Non-Negative Contrastive Learning*. ICML

2026 (published).

XAI/Representation angle: proposed Bayesian gating for non-negative contrastive learning to suppress background/common factors, mitigate optimization conflicts, and improve the interpretability of learned representations.

Keywords: Contrastive Learning, Bayesian Gating, Non-Negative Representations, Interpretability.

Submitted

- **Jiahao Zhang***, Yueying Hu*, Xinyue Xu, Lijie Hu[†]. *Rethinking Circuit Discovery with Learnable Corruptions*. Submitted to NeurIPS 2026.

Mechanistic interpretability angle: introduced SEED, a framework that jointly optimizes semantically admissible corruptions and shared sparse circuits, improving circuit recovery over fixed-corruption attribution baselines across common circuit-discovery tasks.

- **Jiahao Zhang**, Wenshuo Dong, Jie Li, Wei Tang, Lijie Hu[†]. *Coarse-to-Fine Neuron-Path Discovery for Interpretable Vision Transformers*. Submitted to NeurIPS 2026.

Understanding angle: proposed concept-guided coarse-to-fine neuron-path discovery for ViTs, reducing search cost while preserving neuron-level grounding and revealing interpretable cross-layer semantic trajectories.

- Honglang Lu, **Jiahao Zhang**, Xingxiang Li, Jing Li, Yishuo Huang, Xiaoqin Zhong, Lijie Hu, Jun Ma, Zongyao Zhou[†]. *Solar-driven Evapofiltration Enables Co-production of Lithium and Freshwater from Extreme Brines*. Submitted to **Nature Sustainability**.

AI4Science angle: developed a solar-driven photothermal evapofiltration strategy for extreme brines, enabling coupled lithium recovery and freshwater production through brine chemistry and membrane-process design.

Keywords: Brine Chemistry, Lithium Recovery, Solar Energy, Photothermal Membrane, Evapofiltration, AI for Science.

- Haodong Wu, **Jiahao Zhang**, Lijie Hu, Yongqi Zhang[†]. *From Instance Selection to Fixed-Pool Data Recipe Search for Supervised Fine-Tuning*. Submitted to NeurIPS 2026.

Data-centric AI angle: formulated SFT data selection as fixed-pool data recipe search and proposed AutoSelection, a closed-loop framework that searches over filtering, mixing, and deduplication operators to construct stronger instruction-tuning datasets under limited evaluation budgets.

- Xijie Gong, Tingxu Han, **Jiahao Zhang**, Wei Song, Ziqi Ding, Hanqi Yan, Youcheng Sun, Lijie Hu[†]. *How Do Agentic LLMs Decide to Call Tools? A Scaffold Default Controlled by Suppression*. Submitted to NeurIPS 2026.

Mechanistic interpretability angle: constructed minimal contrastive agentic prompts and traced tool-calling decisions to a compact causal vector, showing that agentic scaffolds induce tool use as a default while analysis verbs suppress this default.

In Preparation

- **Jiahao Zhang**[†], Jerry Wang[†]. *TIC: A Unified Framework for Temporal and Causal Inference in Tumor Microenvironments*. Manuscript in preparation.

Target venue: *Cell Systems* (planned submission).

Code: github.com/cellethology/tic. **AI4Science angle:** a unified pipeline combining trajectory inference and causal analysis to study EMT progression and biomarker interactions in spatial transcriptomics.

Research Experience

- **Laboratory of Cell Ethology, CIS, Westlake University**
Research Assistant, Hangzhou, China

Jan. 2025 – Jul. 2025
Advisor: Dr. Jerry Wang

 - Designed and built **TIC**, an AI4Science pipeline for spatial transcriptomics in tumor microenvironments: LLM-assisted cell annotation, graph representation learning, trajectory inference, pseudotime analysis, and causal analysis for EMT-related interactions.
 - Implemented graph-based trajectory modules and validated scientific findings through downstream analyses on EMT progression and biomarker interactions; released and maintained an open-source package.
- **Representation Learning Lab, Westlake University**
Research Assistant, Hangzhou, China

Jun. 2024 – Dec. 2024

 - Developed a diffusion-inspired generative pipeline for **controllable protein affinity** design; implemented training and fine-tuning to generate high- versus low-affinity candidates under controlled conditions.
 - Framed controllability as a reliability/intervention problem: enforced generation constraints and evaluated with structure-aware signals for downstream scientific utility.
- **AI Lab, University of Wisconsin–Madison**
Research Assistant, Remote

Oct. 2023 – Dec. 2023
Advisor: Prof. Chaowei Xiao

 - Built a benchmark for software vulnerability detection and repair; contributed to metric design, experiments on real-world datasets, and systematic analysis of model failure modes and robustness.

Selected Project

- **Interactive QA for Survey Insights (RAG + Evaluation)**
Data Science Research Intern, Grapedata / UC Berkeley

Jan. 2024 – May 2024

 - Built a RAG-based QA system over B2B survey data; led a two-stage workflow combining retrieval and answer verification/evaluation to improve reliability and reduce hallucination in production settings.
 - Impact: prototype commercialized at more than £10,000 per unit; received the Cloud Computing Application Award from the UCB Data Science Discovery Program.

Honors

- Cloud Computing Application Award — UCB Data Science Discovery Program (May 2024)

- Undergraduate Academic Excellence Scholarship (2023–2024)
- Second Prize — College Student Mathematics Competition, Hubei Division (Mar. 2023)
- Second Prize — Chinese Mathematics Competition (Jan. 2023)
- Third Prize — China Undergraduate Mathematical Contest in Modeling (Sept. 2022)

Technical Skills

- **Programming:** Python, C++, Java **Frameworks/Tools:** PyTorch, Git
- **Research Areas:** Mechanistic Interpretability, Vision Transformers, Concept-based Models & Intervention, Agentic LLMs, Data-centric AI, Protein LMs / Generative Modeling, Graph Representation Learning, Causal/Temporal Inference

Last Updated: May 2026

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研究方向

可解释人工智能 (XAI) 作为双向交互系统: 围绕两个核心问题展开研究: (1) 理解 (*Understanding*) —— 从机制层面与概念层面对模型决策进行解释; (2) 干预 (*Intervention*) —— 通过人类引导实现对模型行为的可控调节。

相关方法在 **AI for Science** 场景中进行验证, 重点关注意义解释的真实性 (faithfulness)、可靠性 (reliability) 以及对下游科学任务的实际价值。

研究兴趣

- **Transformer 的机制可解释性:** 概念/神经元路径发现、电路发现、工具调用机制分析、跨层信息流追踪。
- **人机协同干预与模型控制:** 测试阶段引导、基于概念的干预方法、弱监督/间接监督条件下的鲁棒性与可靠性。
- **数据中心与智能体大模型:** 监督微调数据配方搜索、智能体工具使用机制、复杂 prompt 下的可靠行为建模。
- **面向科学发现的可解释 AI (AI4Science):** 面向蛋白质生成、卤水化学、生物学分析、空间组学建模的可解释与可控方法。

教育背景

- **机器学习博士, 穆罕默德·本·扎耶德人工智能大学 (MBZUAI)** 2025.08 – 至今
阿布扎比, 阿联酋 导师: Hu Lijie
研究主题: 交互式可解释人工智能、机制可解释性与 AI4Science。
- **电子信息学士, 华中科技大学 (985)** 2021.09 – 2025.06
武汉, 中国 GPA: 86/100
- **访问学生, 加州大学伯克利分校** 2024.01 – 2024.05
美国加州伯克利 GPA: 3.57/4.0

学术成果

* 表示共同一作 † 表示通讯作者

已接收 / 已发表论文

- **Zhang Jiahao***, Zhang Zeqing*, Wang Di, Hu Lijie†.
Controlling Repetition in Protein Language Models. ICLR 2026 (已接收, Poster)。
XAI/干预方向: 分析蛋白语言模型中的重复退化问题, 并提出推理阶段控制方法 UCCS, 在降低重复生成的同时保持由 AlphaFold 置信度衡量的下游蛋白可折叠性。

- Cui Peng*, **Zhang Jiahao***, Hu Lijie[†].

Bayesian Gated Non-Negative Contrastive Learning. ICML 2026 (已发表)。

XAI/表示学习方向：提出面向非负对比学习的贝叶斯门控机制，用于抑制背景因子与共有因子的干扰，缓解优化冲突，并提升学习表示的可解释性。

关键词： Contrastive Learning, Bayesian Gating, Non-Negative Representations, Interpretability.

已投稿 / 投递中论文

- **Zhang Jiahao***, Hu Yueying*, Xu Xinyue, Hu Lijie[†].

Rethinking Circuit Discovery with Learnable Corruptions. Submitted to NeurIPS 2026.

机制可解释性方向：提出 SEED 框架，在语义可接受的 corruption 空间中联合优化样本级 corruption 与共享稀疏电路，相比固定 corruption 的归因基线，在多个常见电路发现任务上提升电路恢复效果。

- **Zhang Jiahao**, Dong Wenshuo, Li Jie, Tang Wei, Hu Lijie[†].

Coarse-to-Fine Neuron-Path Discovery for Interpretable Vision Transformers. Submitted to NeurIPS 2026.

理解方向：提出面向 ViT 的概念引导粗到细神经元路径发现方法，在保留神经元级 grounding 的同时降低搜索成本，并揭示可解释的跨层语义轨迹。

- Lu Honglang, **Zhang Jiahao**, Li Xingxiang, Li Jing, Huang Yishuo, Zhong Xiaoqin, Hu Lijie, Ma Jun, Zhou Zongyao[†].

Solar-driven Evapofiltration Enables Co-production of Lithium and Freshwater from Extreme Brines. Submitted to Nature Sustainability.

AI4Science 方向：面向极端卤水体系，设计太阳能驱动的光热蒸发过滤策略，通过卤水化学与膜过程设计实现锂资源回收与淡水生产的协同产出。

关键词： Brine Chemistry, Lithium Recovery, Solar Energy, Photothermal Membrane, Evapofiltration, AI for Science.

- Wu Haodong, **Zhang Jiahao**, Hu Lijie, Zhang Yongqi[†].

From Instance Selection to Fixed-Pool Data Recipe Search for Supervised Fine-Tuning. Submitted to NeurIPS 2026.

数据中心 AI 方向：将监督微调数据选择从实例排序扩展为固定数据池上的数据配方搜索，并提出 AutoSelection 闭环框架，在有限评估预算下搜索过滤、混合与去重等数据算子，以构建更高质量的指令微调数据集。

- Gong Xijie, Han Tingxu, **Zhang Jiahao**, Song Wei, Ding Ziqi, Yan Hanqi, Sun Youcheng, Hu Lijie[†].

How Do Agentic LLMs Decide to Call Tools? A Scaffold Default Controlled by Suppression. Submitted to NeurIPS 2026.

机制可解释性方向：构造最小对比式 agentic prompt，并将工具调用决策追踪到一个紧凑的因果向量；结果表明，agentic scaffold 会使工具调用成为默认行为，而分析类动词会抑制这一默认状态。

撰写中 / 准备中论文

- **Zhang Jiahao**[†], Jerry Wang[†].

TIC: A Unified Framework for Temporal and Causal Inference in Tumor Microenvironments. Manuscript in preparation.

拟投期刊: *Cell Systems*。

代码: github.com/cellethology/tic。 **AI4Science 方向:** 提出结合轨迹推断与因果分析的统一流程, 用于研究空间转录组中 EMT 进程与生物标志物交互机制。

科研经历

- **西湖大学细胞行为实验室**

科研助理, 杭州, 中国

2025.01 – 2025.07

导师: Dr. Jerry Wang

 - 设计并构建 TIC 空间转录组分析流程: LLM 辅助细胞注释、图表示学习、伪时间轨迹推断与因果分析。
 - 开发图结构轨迹模块并验证 EMT 相关科学发现; 发布并维护开源工具包。
- **西湖大学表示学习实验室**

科研助理, 杭州, 中国

2024.06 – 2024.12

 - 构建基于扩散模型的可控蛋白亲和力生成框架, 完成训练与微调以生成高/低亲和力候选蛋白。
 - 从“可靠性/干预”角度建模可控生成问题, 并结合结构信号进行下游科学任务评估。
- **University of Wisconsin–Madison AI Lab**

科研助理, 远程

2023.10 – 2023.12

导师: Prof. Chaowei Xiao

 - 构建软件漏洞检测与修复基准, 参与真实数据实验、指标设计与模型鲁棒性分析。

项目经历

- **Survey 数据交互问答系统 (RAG + 评估框架)**

UC Berkeley / Grapedata 数据科学项目实习

2024.01 – 2024.05

 - 构建基于 RAG 的 B2B 调研问答系统, 设计“检索 + 答案验证”双阶段流程, 以提升生产环境下回答的可靠性并降低幻觉。
 - 项目实现商业化, 单套价格超过 £10,000; 获 UCB Data Science Discovery Program Cloud Computing Application Award。

荣誉奖项

- UCB Cloud Computing Application Award (2024)
- 本科优秀奖学金 (2023–2024)
- 全国大学生数学竞赛湖北赛区二等奖 (2023)
- 中国数学竞赛二等奖 (2023)
- 全国大学生数学建模竞赛三等奖 (2022)

技术能力

- **编程语言:** Python, C++, Java

- **框架工具:** PyTorch, Git
- **研究方向:** Transformer 机制可解释性、视觉 Transformer、概念模型与干预、智能体大模型、数据中心 AI、蛋白语言模型/生成建模、图表示学习、因果与时序推断

最后更新: 2026 年 5 月